

clickhouse

S3 NYC

Tutorial

[Advanced Tutorial](#)

Imports

```
# Import jupysql Jupyter extension to create SQL cells
%load_ext sql
%config SqlMagic.autocommit=False
```

Conexión celdas mágicas

```
from getpass import getpass
from os import environ
pwd = getpass()
environ["DATABASE_URL"] = f"clickhouse://{santiago:{pwd}@clickhouse:8123/ibd00"
```

```
%sql
```

DDL

```
%%sql
CREATE TABLE trips
(
    `trip_id` UInt32,
    `vendor_id` Enum8('1' = 1, '2' = 2, '3' = 3, '4' = 4, 'CMT' = 5, 'VTS' =
6, 'DDS' = 7, 'B02512' = 10, 'B02598' = 11, 'B02617' = 12, 'B02682' = 13,
'B02764' = 14, '' = 15),
    `pickup_date` Date,
    `pickup_datetime` DateTime,
    `dropoff_date` Date,
    `dropoff_datetime` DateTime,
    `store_and_fwd_flag` UInt8,
    `rate_code_id` UInt8,
    `pickup_longitude` Float64,
    `pickup_latitude` Float64,
    `dropoff_longitude` Float64,
    `dropoff_latitude` Float64,
    `passenger_count` UInt8,
```

```

        `trip_distance` Float64,
        `fare_amount` Float32,
        `extra` Float32,
        `mta_tax` Float32,
        `tip_amount` Float32,
        `tolls_amount` Float32,
        `ehail_fee` Float32,
        `improvement_surcharge` Float32,
        `total_amount` Float32,
        `payment_type` Enum8('UNK' = 0, 'CSH' = 1, 'CRE' = 2, 'NOC' = 3, 'DIS' =
4),
        `trip_type` UInt8,
        `pickup` FixedString(25),
        `dropoff` FixedString(25),
        `cab_type` Enum8('yellow' = 1, 'green' = 2, 'uber' = 3),
        `pickup_nyct2010_gid` Int8,
        `pickup_ctlabel` Float32,
        `pickup_borocode` Int8,
        `pickup_ct2010` String,
        `pickup_boroc2010` String,
        `pickup_cdeligibil` String,
        `pickup_ntacode` FixedString(4),
        `pickup_ntaname` String,
        `pickup_puma` UInt16,
        `dropoff_nyct2010_gid` UInt8,
        `dropoff_ctlabel` Float32,
        `dropoff_borocode` UInt8,
        `dropoff_ct2010` String,
        `dropoff_boroc2010` String,
        `dropoff_cdeligibil` String,
        `dropoff_ntacode` FixedString(4),
        `dropoff_ntaname` String,
        `dropoff_puma` UInt16
    )
ENGINE = MergeTree
PARTITION BY toYYYYMM(pickup_date)
ORDER BY pickup_datetime;

```

DML

```

%%sql
INSERT INTO trips
SELECT * FROM s3(
    'https://datasets-documentation.s3.eu-west-3.amazonaws.com/nyc-
taxi/trips_{1..2}.gz',
    'TabSeparatedWithNames', ''
    `trip_id` UInt32,
    `vendor_id` Enum8('1' = 1, '2' = 2, '3' = 3, '4' = 4, 'CMT' = 5, 'VTS' =
6, 'DDS' = 7, 'B02512' = 10, 'B02598' = 11, 'B02617' = 12, 'B02682' = 13,
'B02764' = 14, '' = 15),
    `pickup_date` Date,
    `pickup_datetime` DateTime,
    `dropoff_date` Date,
    `dropoff_datetime` DateTime,
    `store_and_fwd_flag` UInt8,

```

```

        `rate_code_id` UInt8,
        `pickup_longitude` Float64,
        `pickup_latitude` Float64,
        `dropoff_longitude` Float64,
        `dropoff_latitude` Float64,
        `passenger_count` UInt8,
        `trip_distance` Float64,
        `fare_amount` Float32,
        `extra` Float32,
        `mta_tax` Float32,
        `tip_amount` Float32,
        `tolls_amount` Float32,
        `ehail_fee` Float32,
        `improvement_surcharge` Float32,
        `total_amount` Float32,
        `payment_type` Enum8('UNK' = 0, 'CSH' = 1, 'CRE' = 2, 'NOC' = 3, 'DIS' =
4),
        `trip_type` UInt8,
        `pickup` FixedString(25),
        `dropoff` FixedString(25),
        `cab_type` Enum8('yellow' = 1, 'green' = 2, 'uber' = 3),
        `pickup_nyct2010_gid` Int8,
        `pickup_ctlabel` Float32,
        `pickup_borocode` Int8,
        `pickup_ct2010` String,
        `pickup_boroct2010` String,
        `pickup_cdeligibil` String,
        `pickup_ntacode` FixedString(4),
        `pickup_ntaname` String,
        `pickup_puma` UInt16,
        `dropoff_nyct2010_gid` UInt8,
        `dropoff_ctlabel` Float32,
        `dropoff_borocode` UInt8,
        `dropoff_ct2010` String,
        `dropoff_boroct2010` String,
        `dropoff_cdeligibil` String,
        `dropoff_ntacode` FixedString(4),
        `dropoff_ntaname` String,
        `dropoff_puma` UInt16
    ") SETTINGS input_format_try_infer_datetimes = 0

```

Contar filas

```
%sql SELECT count() FROM trips limit 5;
```



count()

1999657

SELECT DISTINCT



```
%sql SELECT DISTINCT(pickup_ntaname) FROM trips limit 5;
```

pickup_ntaname
West Village
East Village
Gramercy
Midtown-Midtown South
Yorkville

Echar cuentas

```
%sql SELECT round(avg(tip_amount), 2) FROM trips
```



round(avg(tip_amount), 2)
1.68

GROUP BYs

```
%%sql
SELECT
    passenger_count,
    ceil(avg(total_amount), 2) AS average_total_amount
FROM trips
GROUP BY passenger_count
```



passenger_count	average_total_amount
0	22.69
1	15.97
2	17.15
3	16.76

passenger_count	average_total_amount
4	17.33
5	16.35
6	16.04
7	59.8
8	36.41
9	9.81

```
%%sql
SELECT
    pickup_date,
    pickup_ntaname,
    SUM(1) AS number_of_trips
FROM trips
GROUP BY pickup_date, pickup_ntaname
ORDER BY pickup_date ASC
limit 5;
```

pickup_date	pickup_ntaname	number_of_trips
2015-07-01	Bushwick North	2
2015-07-01	Brighton Beach	1
2015-07-01	Briarwood-Jamaica Hills	3
2015-07-01	Williamsburg	1
2015-07-01	Queensbridge-Ravenswood-Long Island City	9